

## 5.2.49

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# Question

Find all circles that touch the  $x$ -axis at a distance 3 from the origin and have a  $y$ -axis intercept of length  $2\sqrt{7}$ .

# Solution

$$\mathbf{x}^\top \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad \mathbf{C} = -\mathbf{u}, \quad r^2 = \mathbf{C}^\top \mathbf{C} - f \quad (1)$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2)$$

$$\mathbf{C} = \begin{pmatrix} \mathbf{e}_1^\top \mathbf{C} \\ \mathbf{e}_2^\top \mathbf{C} \end{pmatrix} \quad (3)$$

Substituting y-axis in circle equation

$$(\mathbf{y}\mathbf{e}_2)^\top (\mathbf{y}\mathbf{e}_2) + 2\mathbf{u}^\top (\mathbf{y}\mathbf{e}_2) + f = 0 \quad (4)$$

$$\implies y^2 - 2(\mathbf{C}^\top \mathbf{e}_2)y + f = 0 \quad (5)$$

Here the square of Intercept equals to determinant of (5)

$$\Delta = [-2(\mathbf{C}^\top \mathbf{e}_2)]^2 - 4f = 4[(\mathbf{C}^\top \mathbf{e}_2)^2 - f] = (2\sqrt{7})^2 \quad (6)$$

$$(\mathbf{C}^\top \mathbf{e}_2)^2 - f = 7 \quad (7)$$

# Solution

$$\mathbf{C}^\top \mathbf{C} - f = r^2 \quad (8)$$

$$(8) - (7) \implies \mathbf{C}^\top \mathbf{C} - (\mathbf{C}^\top \mathbf{e}_2)^2 = r^2 - 7 \implies (\mathbf{e}_1^\top \mathbf{C})^2 = r^2 - 7 \quad (9)$$

Since Circle touches x-axis at a distance 3 from origin.

$$|\mathbf{e}_2^\top \mathbf{C}| = r \quad (10)$$

$$|\mathbf{e}_1^\top \mathbf{C}| = 3 \implies r^2 = 16 \quad (11)$$

$$(\mathbf{e}_2^\top \mathbf{C})^2 - f = 7, \implies f = 9 \quad (12)$$

$$\mathbf{C} = \begin{pmatrix} \pm 3 \\ \pm 4 \end{pmatrix} \quad (13)$$

$$\mathbf{x}^\top \mathbf{x} + 2(-\mathbf{C})^\top \mathbf{x} + f = 0 \quad (14)$$

$$\mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} -3 & -4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad \mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} -3 & 4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad (15)$$

$$\mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad \mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} 3 & 4 \end{pmatrix} \mathbf{x} + 9 = 0 \quad (16)$$

# C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    double a = 3.0;
    double L = 2.0 * sqrt(7.0);
    double r = sqrt(a*a + (L/2)*(L/2));
    double f = a*a + (L/2)*(L/2) - r*r;

    double centers[4][2] = {
        { a, r},
        { a, -r},
        {-a, r},
        {-a, -r}
    };
};
```

```
for(int i=0; i<4; i++) {  
    double h = centers[i][0];  
    double k = centers[i][1];  
    printf(Circle %d: Centre = (%.2f, %.2f), Radius = %.2f\n,  
           i+1, h, k, r);  
}  
  
printf(Constant term f = %.2f\n, f);  
return 0;  
}
```

# Python Direct

```
import math
import numpy as np
import matplotlib.pyplot as plt
from funcs import circ_gen, circ_param

a = 3.0
L = 2.0 * math.sqrt(7.0)
half = L / 2.0
r = math.sqrt(a*a + half*half)
f = a*a

centers = [
    np.array([[ a],[ r]], dtype=float),
    np.array([[ a],[ -r]], dtype=float),
    np.array([[ -a],[ r]], dtype=float),
    np.array([[ -a],[ -r]], dtype=float)
]
```



```
for i, C in enumerate(centers, 1):
    h = float(round(C[0,0],2))
    k = float(round(C[1,0],2))
    R = float(round(r,2))
    print(fCircle {i}: Centre = ({h}, {k}), Radius = {R})

fig, ax = plt.subplots(figsize=(7,7))
ax.set_aspect('equal', adjustable='box')
ax.grid(True, linestyle='--', alpha=0.5)

for i, C in enumerate(centers,1):
    circ = circ_gen(C, r)
    ax.plot(circ[0,:], circ[1,:], linewidth=1.8)
    ax.scatter(float(C[0,0]), float(C[1,0]), color='black', s=25)
    dx = -0.6 if C[0,0] >= 0 else 0.6
    dy = -0.4 if C[1,0] >= 0 else 0.4
    h = float(round(C[0,0],2))
    k = float(round(C[1,0],2))
```

```
ax.text(h + dx, k + dy, f0{i}({h},{k}), fontsize=10, bbox=
    dict(boxstyle=round,pad=0.2, fc=white, alpha=0.8))

for s in (a, -a):
    ax.scatter(s, 0, color='red', s=50, marker='x')
    ax.text(s + 0.25, -0.45, f({s},0), color='red')

ax.axhline(0, color='gray', linewidth=1)
ax.axvline(0, color='gray', linewidth=1)
ax.set_xlim(-8,8)
ax.set_ylim(-8,8)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Circles tangent to x-axis at (3,0) with y-axis
    intercept 27')
plt.show()
```

```
import ctypes
import numpy as np

lib = ctypes.CDLL('./circles.so')

lib.compute_circles.argtypes = [
    ctypes.c_double,
    ctypes.c_double,
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(4,2)),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double)
]

lib.compute_circles.restype = None
```

```
a = 3.0
L = 2*np.sqrt(7)
centers = np.zeros((4,2), dtype=np.float64)
r = ctypes.c_double()
f = ctypes.c_double()

lib.compute_circles(a, L, centers, ctypes.byref(r), ctypes.byref(
    f))

print(Results from shared library:\n)
for i in range(4):
    h, k = centers[i]
    print(fCircle {i+1}: Centre = ({round(h,2)}, {round(k,2)}),
        Radius = {round(r.value,2)})

print(f\nConstant term f = {round(f.value,2)})
```

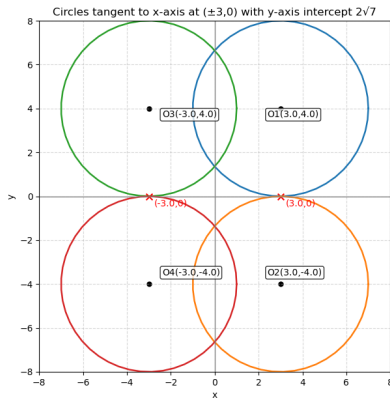


Figure: